

Module 03: Arrays, control structures and Boolean Logic

Objective: Application of arrays, boolean logic and control structures

System requirements: .NET 5.0

Tools: Visual Studio 2022

Duration: ~30min

Author(s): Klaas Hagge & Benedikt Siebert

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1 Leap year calculator

Based on the input of a year, it is to be determined whether it is a leap year or not.

The rules for a leap year are:

- if divisible by 4 => is a leap year
- if also divisible by 100 => is not a leap year
- if also divisible by 400 => is a leap year

The output should look something like this:

```
Enter a year:
2021
2021 is a leap year: false
```

Help:

The modulo operator % can be used to check if a number is divisible by another number.

2 Mini-Lottery

After entering a number between 0 and 100, the user should be told whether he has hit one of the previously defined winning numbers.

To do this, create an Int array and fill it with five numbers between 0 and 100. Then compare the user's input with the array and output the result.

Also check if the submitted number is within the number range between 0 and 100.

```
Enter a number: 99
Congratulations! Your number is one of the five winning numbers!

Enter a number: 4
Your number is not one of the five winning numbers. Better luck next time!
```

Help:

The Contains() function of the array is very helpful here.